Final Exam – Optimization Methods
Yachay Tech University

Estudiante: [Tu Nombre]
Valor de $d = 5$

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Exercise 1 (3 pts)

A company distributes beans and rice. Each sack of rice weighs 10 kg and the sales profit is \$3·d, while each sack of beans weighs 4 kg and the sales profit is \$2·d. The sacks are transported up to a maximum of 450 kg. The transportation labor per sack of beans requires $d + 1$ hours and the transportation labor per sack of rice is d hours. Only 23 hours of labor are available. Determine the number of sacks of beans and rice that should be delivered to maximize the profit.

Formulation of the Linear Programming Problem

Let:

- x_1: number of sacks of rice to deliver
- x_2: number of sacks of beans to deliver

Parameters:

- Each sack of rice weighs 10 kg and brings a profit of \$3·d = \$15
- Each sack of beans weighs 4 kg and brings a profit of \$2·d = \$10
- Maximum transport weight: 450 kg
- Labor:

- Rice: $d = 5$ hours per sack
- Beans: $d + 1 = 6$ hours per sack

- Available labor: 23 hours

Objective function:

$$\text{Maximize } z = 15x_1 + 10x_2$$

Subject to:

$$\begin{aligned} 10x_1 + 4x_2 &\leq 450 && \text{(Weight constraint)} \\ 5x_1 + 6x_2 &\leq 23 && \text{(Labor constraint)} \\ x_1, x_2 &\geq 0 && \text{(Non-negativity)} \end{aligned}$$

Graphical Representation and Integer Solutions

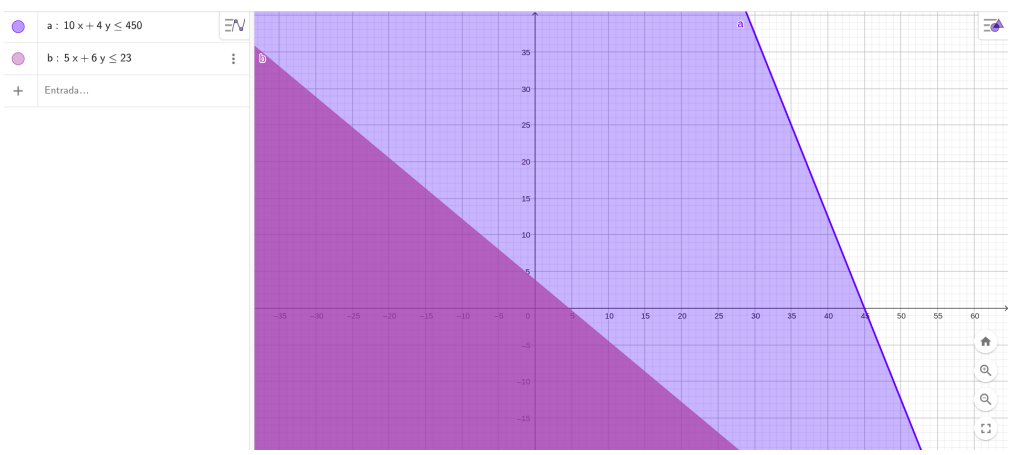
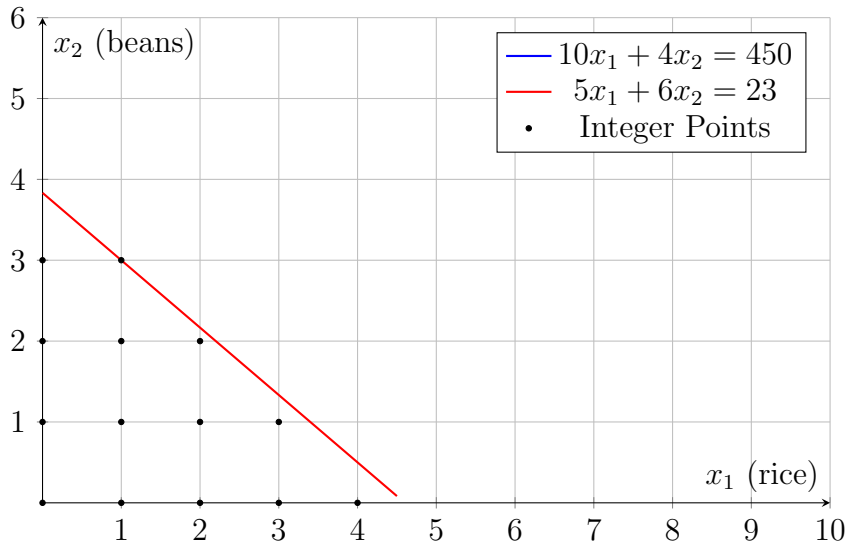


Figure 1: Graphical representation using GeoGebra



Optimal solution using the tableu

We convert the inequalities into equalities by adding slack variables S_1 and S_2:

$$10x_1 + 4x_2 + S_1 = 450$$
$$5x_1 + 6x_2 + S_2 = 23$$

Initial Tableau

Base	x_1	x_2	S_1	S_2	RHS	ratio
S_1	10	4	1	0	450	$450/10 = 45$
S_2	5	6	0	1	23	$23/5 = 4.6$
z	-15	-10	0	0	0	

Entering variable: x_1 (most negative in z-row)

Leaving variable: S_2 (minimum ratio = 4.6)

Pivot: Normalize row S_2

$$\text{New } S_2 = \frac{1}{5}(5,\ 6,\ 0,\ 1,\ 23) = (1,\ \tfrac{6}{5},\ 0,\ \tfrac{1}{5},\ \tfrac{23}{5})$$

Update row S_1:

$$\text{Old } S_1 = (10,\ 4,\ 1,\ 0,\ 450)$$
$$\text{New } S_1 = \text{Old } S_1 - 10 \cdot \text{New } S_2 = (0,\ -8,\ 1,\ -2,\ 404)$$

Update row z:

$$\text{Old } z = (-15,\ -10,\ 0,\ 0,\ 0)$$
$$\text{New } z = \text{Old } z + 15 \cdot \text{New } S_2 = (0,\ 8,\ 0,\ 3,\ 69)$$

Updated Tableau

Base	x_1	x_2	S_1	S_2	RHS	ratio
S_1	0	-8	1	-2	404	—
x_1	1	6/5	0	1/5	23/5	—
z	0	8	0	3	69	

Optimality

Since all coefficients in the z-row are non-negative, the current solution is optimal.

$$x_1 = \frac{23}{5} = 4.6$$
$$x_2 = 0$$
$$z = 69$$

However, since the problem requires integer solutions, we evaluate nearby integer feasible points manually. The best integer solution is verified graphically and by enumeration

Evaluation to find the optimal integer solution

Since the number of sacks must be whole numbers , the problem belongs to the class of Integer Linear Programming . Although the simplex method is a powerful tool for solving linear programming problems, it typically provides solutions with real (fractional)values. Therefore, in this case we opt for a graphical and discrete evaluation of integer points within the feasible region to ensure we obtain a valid integer solution that maximizes the profit.

We evaluate integer combinations (x_1, x_2) satisfying:

$$\begin{cases} 10x_1 + 4x_2 \leq 450 \\ 5x_1 + 6x_2 \leq 23 \end{cases}$$

x_1	x_2	Weight (kg)	Labor (h)	Profit ($)
0	0	0	0	0
1	0	10	5	15
2	0	20	10	30
3	0	30	15	45
4	**0**	**40**	**20**	**60**
0	1	4	6	10
1	1	14	11	25
2	1	24	16	40
3	1	34	21	55
0	2	8	12	20
1	2	18	17	35
2	2	28	22	50
0	3	12	18	30
1	3	22	23	45

This solution maximizes the profit without exceeding weight or labor limitations.

$$\boxed{x_1 = 4, \quad x_2 = 0, \quad z = 15 \cdot 4 + 10 \cdot 0 = 60}$$

Conclusion: The optimal integer solution is to deliver 4 sacks of rice and 0 of beans, with a profit of $60.

Problem

According to the assigned digit $d = 5$, we solve the following Integer Linear Programming Problem using the **Branch and Bound** method.

Objective function:

$$\text{Maximize } z = 3x_1 + 2x_2$$

Subject to:

$$\begin{aligned} x_1 &\leq 5 \\ x_2 &\leq 2d + 1 = 11 \\ x_1 + x_2 &\leq 3d + 0.5 = 15.5 \\ x_1, x_2 &\geq 0, \quad x_1, x_2 \in Z \end{aligned}$$

Step 1: Solve LP Relaxation

We begin by relaxing the integer constraints ($x_1, x_2 \in R$). Evaluating the objective function at key vertices of the feasible region:

- $(0, 0) \Rightarrow z = 0$
- $(0, 11) \Rightarrow z = 2(11) = 22$
- $(5, 10.5) \Rightarrow z = 3(5) + 2(10.5) = 15 + 21 = 36$
- $(5, 0) \Rightarrow z = 3(5) = 15$

The optimal solution to the LP relaxation is:

$$(x_1, x_2) = (5, 10.5), \quad z = 36$$

Since $x_2 = 10.5$ is not integer, we proceed with the branch and bound method.

Step 2: Branch and Bound

Branch 1: $x_2 \leq 10$

This creates the new feasible region:

$$\begin{aligned} x_1 &\leq 5 \\ x_2 &\leq 10 \\ x_1 + x_2 &\leq 15.5 \end{aligned}$$

Evaluating the point $(5, 10)$:

$$z = 3(5) + 2(10) = 15 + 20 = 35$$

This is a feasible **integer** solution.

Branch 2: $x_2 \geq 11$

From the constraint $x_1 + x_2 \leq 15.5$, we get:

$$x_1 \leq 15.5 - 11 = 4.5 \Rightarrow x_1 = 4 \text{ (integer)}$$

Evaluating point $(4, 11)$:

$$z = 3(4) + 2(11) = 12 + 22 = 34$$

This is an integer solution, but worse than the one from Branch 1. Hence, we discard this branch.

Conclusion

The optimal integer solution is:

$$\boxed{(x_1^*, x_2^*) = (5, 10), \quad z^* = 35}$$